

FYP2 Implementation Master Plan

Aspect-Based Sentiment Analysis with Manglish Support

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Version 3 — Model training complete. Binary classification adopted. All three stages confirmed.

1. Project Overview

FYP2 is graded independently of FYP1. Examiners evaluate solely on whether the implemented system achieves the research objectives and answers the research questions. Methods may differ from FYP1 — what matters is that the final system is coherent, justified, and working.

1.1 Core Novelty

The project investigates whether Aspect-Based Sentiment Analysis (ABSA) can be performed effectively using a hybrid pipeline combining traditional dependency grammar rules with classical machine learning — without requiring fine-tuned transformer or BERT-based models. Extended in three directions:

- A hybrid rule-ML pipeline extracting aspect-opinion pairs and classifying binary sentiment (Positive / Negative) using an SVM-based ensemble.
- Manglish support through a curated slang normalisation dictionary and machine translation fallback.
- An interpretable Streamlit dashboard turning raw sentiment outputs into visual analytics for business stakeholders.

1.2 Binary Classification Decision — Formally Adopted

The system performs binary sentiment classification (Positive / Negative only). The neutral class is excluded. This decision was made before model training and is justified on three grounds:

Ground 1 — Empirical observation of neutral labels

Manual inspection of SemEval 2014 Restaurant neutral-labeled rows revealed annotation artefacts — aspects mentioned without direct sentiment expression — not genuine customer neutrality. Examples: 'my wife and I had a wonderful experience' labeled neutral for aspect 'dinner'; 'waiting 20 minutes for drinks, rude bartender' labeled neutral for aspect 'drinks'.

Ground 2 — Statistical sparsity and linguistic incoherence

Neutral accounts for 17.7% of training rows and 6.6% of test rows. Unlike negative reviews (which consistently use negative language), neutral reviews contain a mixture of positive, negative, and objective language. Sparsity plus incoherence together justify removal.

Ground 3 — Literature precedent

Xinzhai Ai et al. (Multi-Layer Attention Approach for ABSA) explicitly tested bi-class versus tri-class and found neutral labels 'greatly affects the accuracy of the model.' Bi-class accuracy improved from 80.6% to 91.0% on the restaurant subset.

1.3 Confidence Threshold

At inference time, predictions with winning-class probability < 0.65 are flagged as low-confidence on the dashboard UI. No neutral label is introduced. The flag communicates model uncertainty, not a third sentiment class. In testing, 8.3% of predictions (70/840) fell below this threshold.

2. Research Objectives and Questions

RO	Objective
RO1	Develop a hybrid rule-based and machine learning model that builds upon traditional sentiment analysis to improve contextual understanding of customer feedback.
RO2	Design and train a sentiment classification mechanism capable of identifying and categorising distinct aspects within customer reviews with high accuracy.
RO3	Develop an interactive visualisation dashboard that presents analysed feedback data in a structured format to provide clear insights into customer sentiment trends.
RO4	Extend the framework to support both English and Malay language inputs, accommodating the multilingual feedback context found in Malaysia.

RQ	Question
RQ1	How effective is a hybrid rule-based and machine learning approach in extracting specific aspects from unstructured customer feedback?
RQ2	To what extent can the proposed model accurately classify sentiments across multiple distinct aspects within a single review?
RQ3	How can aspect-based sentiment data be effectively visualised to provide actionable insights for business stakeholders?
RQ4	What is the feasibility of extending the sentiment analysis framework to process bilingual (English and Malay) inputs accurately?

3. Confirmed File Structure

Component	Status	Notes
notebooks/01_data_pipeline.ipynb	Done	Local file loading, XML + ASTE parsing, standardisation. Saves Train.csv and Test.csv.
notebooks/02_eda.ipynb	Done	Full EDA. All plots saved to ../figures/.
notebooks/03_model_training.ipynb	Done	Three-stage training complete. All results confirmed on Google Colab.

notebooks/04_dashboard_data_prep.ipynb	Not Started	Processes Malaysian review CSVs offline. Outputs structured CSV for dashboard.
src/normaliser.py	Done	27 tests passing. Loads manglish_dict.json.
src/translator.py	Done	18 tests passing. Returns {original, translated, was_translated}.
src/extractor.py	Done	31 tests passing. 5 rules + 3 helpers + proximity fallback.
src/categoriser.py	Done	27 tests passing. Keyword overrides + similarity check.
src/pipeline.py	Done	30 tests passing. Single run_pipeline() function.
src/utils.py	Done	standardize_text() + standardize_dataframe_aspects().
data/manglish_dict.json	Done	Manglish dictionary as JSON.
data/Train.csv / Test.csv	Done	Clean SemEval 2014 Restaurant data. Train: 3608 rows, Test: 994 rows.
data/train_categorized.csv	Done	3061 rows after categoriser drop (15.2% dropped). Binary training set: 2516 rows after neutral removal.
data/test_categorized.csv	Done	902 rows after categoriser drop. Binary test set: 840 rows after neutral removal.
figures/	Done	All EDA and model training plots saved here.
models/model_proposed.pkl	Done	Stage 3 final model — Binary SVM + Sentence Transformer ensemble.
models/ohe_proposed.pkl	Done	Stage 3 OHE encoder.
models/transformer_reference.txt	Done	Reference: paraphrase-multilingual-MiniLM-L12-v2.
models/model_binary_tfidf.pkl	Done	Stage 2 baseline model — Binary SVM TF-IDF.
models/tfidf_binary.pkl	Done	Stage 2 TF-IDF vectorizer.
models/ohe_binary.pkl	Done	Stage 2 OHE encoder.
dashboard/app.py	Not Started	Streamlit app. Two pages: Inference Sandbox and Brand Analytics Dashboard.
playground.py	Done	Working interactive tester.

4. Confirmed Model Training Results

All three stages trained and evaluated on SemEval 2014 Restaurant data. Stages 2 and 3 trained on Google Colab due to RAM constraints on local machine (8GB with 85% baseline usage). Results confirmed and reproducible.

4.1 Stage 1 — Tri-class SVM Baseline (Historical Reference)

Class	Precision	Recall	F1	Support
negative	0.6949	0.6260	0.6586	131
neutral	0.3108	0.3710	0.3382	62
positive	0.9000	0.9013	0.9006	709
macro avg	—	—	0.6325	902

4.2 Stage 2 — Binary SVM Baseline (Neutral Dropped, TF-IDF)

△ Classifier: SVM (max_iter=1000) + LogisticRegression (liblinear, C=5.0) + ComplementNB. C reduced from 8.08 to 5.0 and solver changed from saga to liblinear due to compute constraints. Performance impact negligible.

Class	Precision	Recall	F1	Support
negative	0.6364	0.6947	0.6642	131
positive	0.9426	0.9267	0.9346	709
macro avg	—	—	0.7994	840

4.3 Stage 3 — Binary SVM + Sentence Transformer (Proposed System)

△ ComplementNB excluded — incompatible with signed continuous transformer embeddings. Replaced with second LogisticRegression (C=1.0, liblinear). SVM convergence warning at max_iter=1000 noted — attempts to resolve via max_iter=2000 and StandardScaler both reduced performance, confirming early-stopped SVM generalises better on this hybrid feature space.

Class	Precision	Recall	F1	Support
negative	0.7597	0.8931	0.8211	131
positive	0.9796	0.9478	0.9634	709
macro avg	—	—	0.8922	840

4.4 Results Progression Summary

Stage	Classification	Features	Macro-F1	Change
Stage 1 — Tri-class SVM	3-class	TF-IDF + sentiment + OHE	0.6325	Baseline

Stage 2 — Binary SVM	2-class	TF-IDF + sentiment + OHE	0.7994	+0.1669 (neutral drop)
Stage 3 — Binary SVM + Transformer	2-class	SentenceTransformer + sentiment + OHE	0.8922	+0.0928 (transformer)
Total improvement	—	—	+0.2597	Baseline → Proposed

5. Ensemble Architecture — Stage 3 vs Stage 2

Component	Stage 2 (TF-IDF)	Stage 3 (Transformer)
Estimator 1	SVC(linear, C=5, max_iter=1000)	SVC(linear, C=5, max_iter=1000)
Estimator 2	LogisticRegression(C=5, liblinear)	LogisticRegression(C=5, liblinear)
Estimator 3	ComplementNB(alpha=0.012)	LogisticRegression(C=1.0, liblinear) — CNB replaced, incompatible with signed embeddings
Voting	Soft	Soft
Feature dims	~8011 (sparse TF-IDF dominant)	395 (384 dense embeddings + 6 sentiment + 5 OHE)
Resampling	BorderlineSMOTE — 1834 per class	BorderlineSMOTE — 1834 per class

6. Known Limitations and Error Analysis Findings

Limitation	Detail	Where Documented
Ambiance and price negative F1 degraded at inference	Only 58 and 81 real negative training examples respectively. SMOTE generated synthetic samples in ambiguous embedding space. Model defaults to near-random confidence for these categories.	Error analysis section
SVM convergence warning at max_iter=1000	SVM terminated early in Stage 3. Increasing to 2000 and applying StandardScaler both reduced test performance — early stopping acts as implicit regularisation on hybrid dense/sparse feature matrix.	Model training section
ComplementNB excluded from Stage 3	CNB requires non-negative input. Transformer embeddings contain negative values. Replaced with second LR estimator.	Methodology section

Extraction engine misses multi-word Malay food terms	spaCy en_core_web_md fails to parse 'nasi lemak', 'mee goreng' etc. as compound nouns reliably.	Error analysis
Distribution shift	Train: 60% positive. Test: 84.4% positive (after neutral removal). BorderlineSMOTE on train only.	Limitations chapter
Single annotator for Malaysian gold standard	150 rows manually verified by one annotator. LLM agreement rate used as external consistency check.	Limitations chapter
SVM hyperparams changed from Optuna best	Original Optuna best: C=8.08, saga. Changed to C=5.0, liblinear due to compute constraints. Convergence evidence suggests minimal performance impact.	Methodology note

7. Benchmark and Evaluation Framework

7.1 Internal Progression (Primary Comparison)

The three-stage internal progression is the primary evidence in the results chapter. It isolates the effect of each decision: neutral drop (Stage 1→2) and feature extractor upgrade (Stage 2→3).

7.2 External Literature Context (Tri-class — Not Directly Comparable)

Paper	Method	Metric	Score
Li et al. 2022 — PMHSAT-BiGRU	BiGRU + multi-head self-attention. No pretrained LM.	Macro-F1 (3-class)	69.25%
Sharma & Kaur 2021 — ASGCN	Aspect-Specific Graph CNN.	Macro-F1 (3-class)	73.76%
Yang & Zeng 2021 — LCA-MHSA	Local context-aware + multi-head self-attention. No BERT.	Macro-F1 (3-class)	73.97%
Phan & Ogunbona 2020 — LCFS-ASC-CDW	Local Context Focus + BERT/RoBERTa.	Macro-F1 (3-class)	80.31%
Silva & Marcacini 2021 — ABSA-DeBERTa	Disentangled attention within DeBERTa.	Macro-F1 (3-class)	86.11%

⚠ Direct numeric comparison between binary and 3-class results is NOT made in the report. External papers cited as landscape context only.

7.3 Binary Classification Precedent

Paper	Finding
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Xinzhi Ai et al. — LSTM-MATT-AC	Explicitly tested bi-class vs tri-class. Found neutral labels degrade accuracy. Bi-class improved from 80.6% to 91.0% on restaurant subset. Cited for methodological justification of neutral exclusion only.
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8. Malaysian Data — Collection and Annotation

Restaurant	Status	Notes
Jom Corner (J Corner)	Confirmed	Strongest Manglish source. Heavy mid-sentence code-switching.
Raihana One Bistro	Confirmed	Rich negative reviews. Good multi-aspect diversity.
Saba Restaurant Cyberjaya	Confirmed	More formal register. Provides contrast.

8.1 Annotation Protocol

- LLM generates candidate labels: aspect_category, opinion_word, sentiment, is_manglish flag.
- Manually verify 50 reviews per restaurant (150 rows total). Researcher judgment is ground truth.
- Record LLM agreement rate — reported as annotation consistency measure.
- Mark 150 verified rows as human_verified=True. Gold Standard test set.
- Remaining rows (human_verified=False) used only for dashboard population.
- Annotation schema: review_id, review_text, restaurant_name, aspect_category (Food/Service/Ambiance/Price/Other), opinion_word, sentiment (Positive/Negative), is_manglish, human_verified.

⚠ Neutral excluded from annotation schema — consistent with binary classification decision.

9. Dashboard

9.1 Page 1 — Inference Sandbox

- Text input for a single review string.
- Toggle: English mode / Manglish mode.
- On submit: original text, normalised text, translated English, extracted aspect-opinion pairs, predicted sentiment per pair (Positive/Negative), aspect category.
- Predictions with confidence < 0.65 flagged visually — amber colour, confidence percentage shown. Clean high-confidence predictions show label only.
- If no aspect-opinion pairs extracted: display 'No aspect-opinion pairs found — cannot process text.'

9.2 Page 2 — Brand Analytics Dashboard

- Restaurant selector dropdown.
- KPI scorecard: Total Reviews Analysed, Brand Health Score (% positive), Most Discussed Aspect.
- Plotly donut chart: binary sentiment distribution (Positive / Negative).
- 100% stacked horizontal bar chart: sentiment breakdown per aspect.
- Positive and negative word clouds side by side.
- Top-10 negative bigrams bar chart.

⚠ All dashboard labels are Positive / Negative only. Neutral removed from all display elements.

10. What Is Done vs What Remains

Component	Status
src/ — all 5 modules, 133 tests passing	Done
notebooks/01_data_pipeline.ipynb	Done
notebooks/02_eda.ipynb	Done
notebooks/03_model_training.ipynb — all 3 stages	Done
Stage 1 tri-class baseline: 0.6325 Macro-F1	Done
Stage 2 binary TF-IDF baseline: 0.7994 Macro-F1	Done
Stage 3 binary transformer proposed system: 0.8922 Macro-F1	Done
All 6 model pickle files saved to Drive and local	Done
figures/ — all training and EDA plots	Done
Extraction engine — multi-word Malay term improvements	Needs work
Manglish dictionary — expand coverage	Needs work
Translation engine — edge case handling	Needs work
Malaysian review scraping (200-300 per restaurant)	Not done
Malaysian annotation (150 verified rows)	Not done
notebooks/04_dashboard_data_prep.ipynb	Not started
dashboard/app.py — Streamlit	Not started

11. Methodology Narrative for Report

The methodology chapter must read as a coherent empirical progression. Recommended narrative arc:

1. Establish the problem: ABSA requires aspect-level feature representation. TF-IDF assigns equal weight to all words regardless of which aspect is being classified — causing positive words to dominate negative aspect classifications in mixed-sentiment sentences.
2. Establish the data decision: Manual inspection of SemEval neutral labels revealed annotation artefacts. Supported by class statistics and Ai et al. literature. Binary classification adopted before any model was trained.
3. Present the pipeline architecture: Rule-based extraction → aspect categorisation → feature encoding → SVM classification. Each stage addresses a specific RO.
4. Present the feature extractor upgrade: paraphrase-multilingual-MiniLM-L12-v2 used as feature extractor only. No fine-tuning. Justified by aspect-level representation problem. Multilingual capability addresses RO4.
5. Present the results progression: Stage 1 → Stage 2 → Stage 3. Each improvement attributed to specific decisions. Total gain of +0.2597 Macro-F1.
6. Contextualise against literature: External benchmarks cited as landscape context only. System contribution is the Manglish pipeline and interpretable architecture, not raw benchmark score.

12. Compute Environment Notes

- Local machine: Asus Vivobook, AMD Ryzen 5 5600H, 8GB RAM. 85% baseline RAM usage makes heavy training impossible without paging.
- Google Colab: Used for all three model training stages. 12.7GB RAM available, 1.8GB used during training. Free tier — runtime may disconnect after inactivity.
- Colab file structure: My Drive/FYP-Colab/Data/, Models/, Figures/, Notebook/, SRC/.
- Key finding: probability=True on SVC without max_iter causes indefinite training on local machine. Fix: max_iter=1000 on all SVC instances.
- Optuna: Stage 1 completed 50 trials. Stage 2 converged at trial 0 (warm start) — interrupted at trial 18. Stage 3 Optuna skipped — convergence evidence sufficient.
- Sentence transformer encoding: ~1m 8s for training set (2516 rows), ~25s for test set (840 rows) on Colab CPU.